



# Carry-less product: `std::clmul`

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**GitHub Issue:**

[wg21.link/P3642/github](https://wg21.link/P3642/github)

**Source:**

[github.com/Eisenwave/cpp-proposals/blob/master/src/clmul.cow](https://github.com/Eisenwave/cpp-proposals/blob/master/src/clmul.cow)

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**ABSTRACT:** Add widening and non-widening carry-less multiplication functions.

## § Revision history

### § Changes since R0

- Generate the proposal using COWEL instead of bikeshed
- Fix incorrect formula in §5. [Proposed wording](#) for bits  $\geq$  the integer width
- Fix §3.1. [Hardware support](#) missing new VPCLMULQDQ instructions
- Fix improper uses of `std::unsigned_integral` in §1. [Introduction](#)
- Make slight editorial wording adjustments
- Rebase on [\[N5008\]](#) and [\[P3161R4\]](#)
- Mention [\[SimdJsonClmul\]](#) in §2. [Motivation](#)

## § Contents

### 1 Introduction



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## § 1. Introduction

**Carry-less multiplication** is a simple numerical operation on unsigned integers. It can be seen as a regular multiplication where **xor** is being used as a reduction instead of **+**.

It is also known as "XOR multiplication" and "polynomial multiplication". The latter name is used because mathematically, it is equivalent to performing a multiplication of two polynomials in GF(2), where each bit is a coefficient.

I propose a `std::clmul` function to perform this operation:

```
template<class T>
constexpr T clmul(T x, T y) noexcept;
```

I also propose a widening operation in the style of [P3161R4], as follows:

```
template<class T>
struct mul_wide_result {
    T low_bits;
    T high_bits;
};

template<class T>
constexpr mul_wide_result<T> clmul_wide(T x, T y) noexcept;
```

## § 2. Motivation

Carry-less multiplication is an important operation in a number of use cases:

- **CRC Computation:** While cyclic redundancy checks can theoretically be performed with a finite field of any length, in practice, **GF(2)[X]**, the *polynomial ring* over the *Galois field*

with two elements is used. Polynomial addition in this ring can be implemented via `xor`, and multiplication via `clmul`, which makes cyclic redundancy checks considerably faster.

- **Cryptography:** `clmul` may be used to implement AES-GCM. [\[IntelCmul\]](#) describes this process in great detail and motivates hardware support for carry-less multiplication via the `pclmulqdq` instruction.
- **Bit manipulation:** `clmul` performs a large amount of `<<` and `xor` operations in parallel. This is utilized in the reference implementation [\[BitPermutations\]](#) of `std::bit_compressor`, proposed in [\[P3104R3\]](#). For example, the form `clmul(x, -1u)` computes the bitwise inclusive parity for each bit of `x` and the bits to its right.

Carry-less multiplication is of such great utility that there is widespread hardware support, some dating back more than a decade. See below for motivating examples.

## § 2.1. Parity computation and JSON parsing

The *parity* of an integer `x` is `0` if the number of one-bits is even, and `1` if it is odd. The parity can also be computed with `popcount(x) & 1`.



**EXAMPLE:** The special form `clmul(x, -1)` computes the parity of each bit in `x` and the bits to its right. The most significant bit holds the parity of `x` as a whole.

```
bool parity(std::uint32_t x) {  
    return std::clmul(x, -1u) >> 31;  
}
```

While the parity of *all* bits can be obtained with `clmul`, it computes the inclusive cumulative parity, which can be used to accelerate parsing JSON and other file formats ([\[SimdJsonCmul\]](#)). This can be done by mapping each `"` character onto a `1`-bit, and any other character onto `0`. `clmul(x, -1)` would then produce masks where string characters corresponds to a `1`-bit.



**EXAMPLE:**

```
abc xxx "foobar" zzz "a" // input string  
00000000010000000100000101 // quote_mask  
000000000.111111.00000.1. // clmul(quote_mask, -1), ignoring 1-bits of quote_mask
```

## § 2.2. Fast space-filling curves

The special form `clmul(x, -1)` can be used to accelerate the computation of Hilbert curves. To properly understand this example, I will explain the basic notion of space-filling curves.

We can fill space using a 2D curve by mapping the index `i` on the curve onto Cartesian coordinates `x` and `y`. A naive curve that fills a `4x4` square can be computed as follows:

```
struct pos { uint32_t x, y; };

pos naive_curve(uint32_t i) { return { i % 4, i / 4 }; }
```



When mapping the index  $i = 0, 1, \dots, 0xf$  onto the returned 2D coordinates, we obtain the following pattern:

```
0 1 2 3
4 5 6 7
8 9 a b
c d e f
```

The problem with such a naive curve is that adjacent indices can be positioned very far apart (the distance increases with row length). For image processing, if we store pixels in this pattern, cache locality is bad; two adjacent pixels can be very far apart in memory.

A **Hilbert curve** is a family of space-filling curves where the distance between two adjacent elements is 1:

```
0 1 e f
3 2 d c
4 7 8 b
5 6 9 a
```

De-interleaving bits of  $i$  into  $x$  and  $y$  yields a **Z-order curve**, and performing further transformations yields a **Hilbert curve**.



**EXAMPLE:** `clmul` can be used to compute the bitwise parity for each bit and the bits to its right, which is helpful for computing Hilbert curves. Note that the following example uses the `std::bit_compress` function from [P3104R3], which may also be accelerated using `clmul`.

```
pos hilbert_to_xy(uint32_t i)
{
    // De-interleave the bits of i.
    uint32_t i0 = std::bit_compress(i, 0x55555555u); // abcdefgh → bdfh
    uint32_t i1 = std::bit_compress(i, 0xaaaaaaaaau); // abcdefgh → aceg

    // Undo the permutation that Hilbert curves apply on top of Z-order
    uint32_t A = i0 & i1;
    uint32_t B = i0 ^ i1 ^ 0xffffu;
    uint32_t C = std::clmul(A, -1u) >> 16;
    uint32_t D = std::clmul(B, -1u) >> 16;

    uint32_t a = C ^ (i0 & D);
    return { .x = a ^ i1, .y = a ^ i0 ^ i1 };
}
```

This specific example is taken from [\[FastHilbertCurves\]](#). [\[HackersDelight\]](#) explains the basis behind this computation of Hilbert curves using bitwise operations.



When working with space-filling curves, the inverse operation is also common: mapping the Cartesian coordinates onto an index on the curve. In the case of Z-order curves aka. Morton curves, this can be done by simply interleaving the bits of  $x$  and  $y$ . A Z-order curve is laid out as follows:

```
0 1 4 5
2 3 6 7
8 9 c d
a b e f
```



**EXAMPLE:** `clmul` can be used to implement bit-interleaving in order to generate a Z-order curves.

```
uint32_t xy_to_morton(uint32_t x, uint32_t y)
{
    uint32_t lo = std::clmul(x, x) << 0; // abcd -> 0a0b0c0d
    uint32_t hi = std::clmul(y, y) << 1; // abcd -> a0b0c0d0
    return hi | lo;
}
```



**NOTE:** In the example above, `std::clmul(x, x)` is equivalent to [\[P3104R3\]](#)'s `std::bit_expand(x, 0x55555555u)`.

## § 3. Possible implementation

A naive and unconstrained implementation looks as follows:

```
template<class T>
constexpr T clmul(const T x, const T y) noexcept
{
    T result = 0;
    for (int i = 0; i < numeric_limits<T>::digits; ++i) {
        result ^= (x << i) * ((y >> i) & 1);
    }
    return result;
}
```

### § 3.1. Hardware support

The implementation difficulty lies mostly in utilizing available hardware instructions, not in the naive fallback implementation.

In the following table, let  $uN$  denote  $N$ -bit unsigned integer operands, and  $\times N$  denote the amount of operands that are processed in parallel.



| Operation                         | x86_64                                          | ARM                                       | RV64                                           |
|-----------------------------------|-------------------------------------------------|-------------------------------------------|------------------------------------------------|
| <code>clmul u64×4 → u128×4</code> | <code>vpcldmulqddq</code> <sup>VPCLMULQDQ</sup> |                                           |                                                |
| <code>clmul u64×2 → u128×2</code> | <code>vpcldmulqddq</code> <sup>VPCLMULQDQ</sup> |                                           |                                                |
| <code>clmul u64 → u128</code>     | <code>pclmulqddq</code> <sup>PCLMULQDQ</sup>    | <code>pmull+pmull2</code> <sup>Neon</sup> | <code>clmul+clmulh</code> <sup>Zbc, Zbkc</sup> |
| <code>clmul u64 → u128</code>     | <code>pclmulqddq</code> <sup>PCLMULQDQ</sup>    | <code>pmull+pmull2</code> <sup>Neon</sup> | <code>clmul+clmulh</code> <sup>Zbc, Zbkc</sup> |
| <code>clmul u64 → u64</code>      |                                                 | <code>pmull</code> <sup>Neon</sup>        | <code>clmul</code> <sup>Zbc, Zbkc</sup>        |
| <code>clmul u8×8 → u16×8</code>   |                                                 | <code>pmull</code> <sup>Neon</sup>        |                                                |
| <code>clmul u8×8 → u8×8</code>    |                                                 | <code>pmul</code> <sup>Neon</sup>         |                                                |



**EXAMPLE:** A limited x86\_64 implementation of `clmul_wide` may look as follows:

```
#include <immintrin.h>
#include <stdint.h>

mul_wide_result<uint64_t> clmul_wide(uint64_t x, uint64_t y) noexcept
{
    __m128i x_128 = _mm_set_epi64x(0, x);
    __m128i y_128 = _mm_set_epi64x(0, y);
    __m128i result_128 = _mm_clmulepi64_si128(x_128, y_128, 0);
    return {
        .low_bits = uint64_t(_mm_extract_epi64(result_128, 0)),
        .high_bits = uint64_t(_mm_extract_epi64(result_128, 1))
    };
}
```

There also exists an LLVM pull request ([\[LLVMClmul\]](#)) which would add an `@llvm.clmul` intrinsic function.

## § 4. Design considerations

Multiple design choices lean on [\[P0543R3\]](#) and [\[P3161R4\]](#). Specifically,

- the choice of header `<numeric>`,
- the choice to have a widening operation,
- the `_wide` naming scheme,

- the `mul_wide_result` template, and
- the decision to have a `(T, T)` parameter list.



## § 4.1. Naming

Carry-less multiplication is also commonly called "Galois Field Multiplication" or "Polynomial Multiplication".

The name `clmul` was chosen because it carries no domain-specific connotation, and because it is widespread:

- Intel refers to PCLMULQDQ As "Carry-Less Multiplication Quadword" in its manual; see [\[IntelManual\]](#).
- RISC-V refers to `clmul` as carry-less multiplication, and this is obvious from the mnemonic.
- The Wikipedia article ([\[WikipediaClmul\]](#)) for this operation is titled "Carry-less product".
- The (proposed) LLVM intrinsic function ([\[LLVMClmul\]](#)) is tentatively named `@llvm.clmul`.

## § 4.2. SIMD support

The proposal is currently limited to the scalar version of `clmul`. However, `<simd>` support may be useful to add, and would be backed by hardware support to an extent.



**DECISION:** If LEWG approves, should decide whether `<simd>` support should be added in *this* paper, or in a separate paper, if at all.

## § 5. Proposed wording

The proposed changes are relative to the working draft of the standard as of [\[N5008\]](#), with the changes in [\[P3161R4\]](#) applied.

Update subclause [\[version.syn\] paragraph 2](#) as follows:



```
#define __cpp_lib_clamp                201603L // also in <algorithm>
#define __cpp_lib_clmul                20????L // also in <numeric>
[...]
#define __cpp_lib_overflow_arithmetic 20?????L 20?????L // also in <numeric>
```

Update the synopsis in [\[numeric.ops.overview\]](#) as follows:



```
template<class T, class U>
constexpr T saturate_cast(U x) noexcept; // freestanding

template<class T>
```



```
struct add_carry_result { // freestanding
    T low_bits;
    bool overflow;
};

template<class T>
using sub_borrow_result = add_carry_result; // freestanding

template<class T>
struct mul_wide_result { // freestanding
    T low_bits;
    T high_bits;
};

template<class T>
struct div_result { // freestanding
    T quotient;
    T remainder;
};

template<class T>
constexpr add_carry_result<T>
    add_carry(T x, T y, bool carry) noexcept; // freestanding
template<class T>
constexpr sub_borrow_result<T>
    sub_borrow(T left, T right, bool borrow) noexcept; // freestanding
template<class T>
constexpr mul_wide_result<T>
    mul_wide(T x, T y) noexcept; // freestanding
template<class T>
constexpr mul_wide_result<T>
    clmul_wide(T x, T y) noexcept; // freestanding
template<class T>
constexpr div_result<T>
    div_wide(T dividend_high, T dividend_low, // freestanding
             T divisor) noexcept;

template<class T>
constexpr bool
    is_div_defined(T dividend, T divisor) noexcept // freestanding
template<class T>
constexpr bool
    is_div_wide_defined(T dividend_high, // freestanding
                       T dividend_low,
                       T divisor) noexcept;

// [numeric.clmul], carry-less product
template<class T>
constexpr T clmul(T x, U y) noexcept; // freestanding
```

In subclause [\[numeric.overflow\]](#) (known as [\[numeric.sat\]](#) at the time of writing), insert the following item, immediately following the description of `mul_wide`:





```
template<class T>
constexpr mul_wide_result<T> clmul_wide(T x, T y) noexcept;
```



Let  $\oplus$  denote the exclusive OR operation ([[expr.xor](#)]). Given an integer  $\alpha$ , let  $\alpha_i$  denote the  $i^{\text{th}}$  least significant bit in the base-2 representation of  $\alpha$ .

*Constraints:*  $T$  is an unsigned integer type ([[basic.fundamental](#)]).

*Returns:* An object storing the bits of an integer  $c$ , where the value of  $c_i$  is given by Formula [?.?](#),  $x$  is  $x$ ,  $y$  is  $y$ , and  $N$  is the width of  $T$ . The result object is initialized so that

- `low_bits` stores the  $N$  least significant bits of  $c$ , and
- `high_bits` stores the subsequent  $N$  bits of  $c$ .

$$c_i = \bigoplus_{j=0}^i x_j y_{i-j} \quad \text{[FORMULA ??.]}$$



**WARNING:** If the mathematical notation in the block above does not render for you, you are using an old browser with no MathML support. Please open the document in a recent version of Firefox or Chrome.



**NOTE:** The formula is taken from [[IntelCmul](#)], with different variable names, and with no special case for the upper  $N$  bits; we can simply treat the integers as mathematical integers with  $2N$  width.

See [[iterator.concept.wine](#)] for precedent on using  $N$  to denote the width of a type.

See [[sf.cmath.riemann.zeta](#)] for precedent on wording which includes formulae.



**NOTE:** The formula above in TeX notation is:

```
c_i = \bigoplus_{j = 0}^i x_i y_{i - j}
```

In subclause [[numeric.ops](#)], append a subclause immediately following [[numeric.overflow](#)] (known as [[numeric.sat](#)] at the time of writing):



## Carry-less product

[[numeric.clmul](#)]

```
template<class T>
constexpr T clmul(T x, T y) noexcept;
```

*Constraints:*  $T$  is an unsigned integer type ([[basic.fundamental](#)]).

*Returns:* `clmul_wide(x, y).low_bits` ([[numeric.overflow](#)]).



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- [LLVMCmul] *Oscar Smith*. [IR] Add llvm cmul intrinsic  
<https://github.com/llvm/llvm-project/pull/140301>